CSharPN

A Code-First Framework for Coloured Petri Net Modelling with C# as Inscription Language

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PNSE'26 · Petri Nets and Software Engineering · June 2026

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> *What if a CPN model were just a C# file in your repo?*

CPNs in 30 seconds

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- **Timed CPNs**: tokens carry timestamps, global clock advances
- **Hierarchical CPNs**: substitution transitions + port places (In, Out, I/O)

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Open source (MIT) — github.com/simontjell/CSharPN.public

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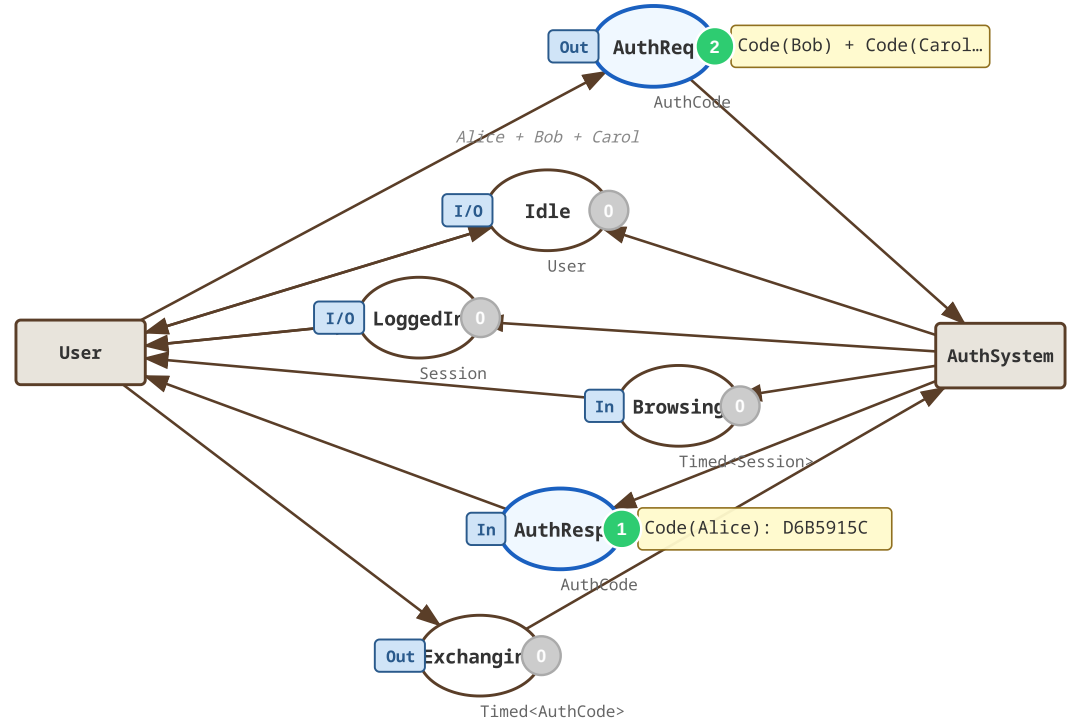
Running example: OAuth2 authoris

Exercises all three features:

Top page: shared places link the two substitution transitions.



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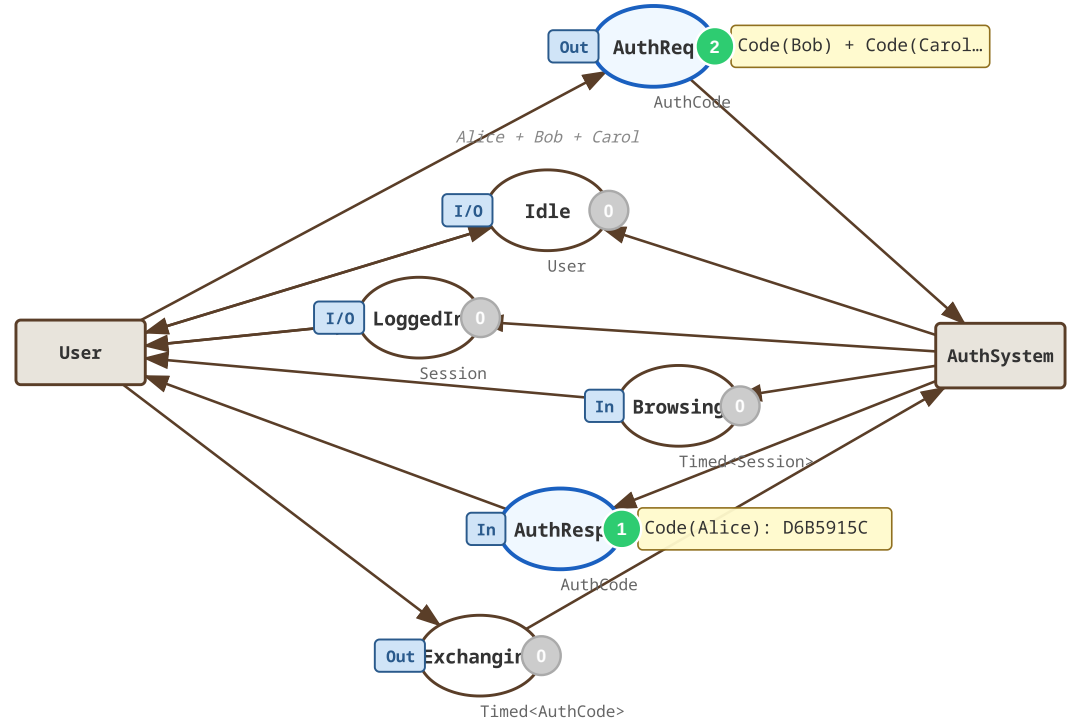

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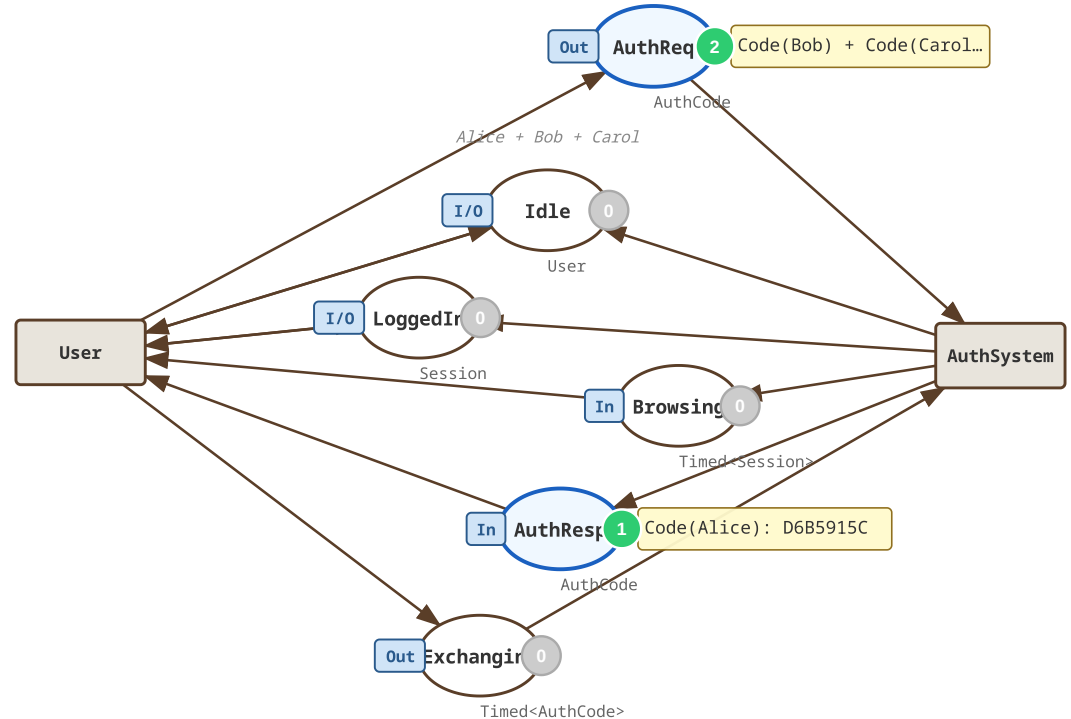

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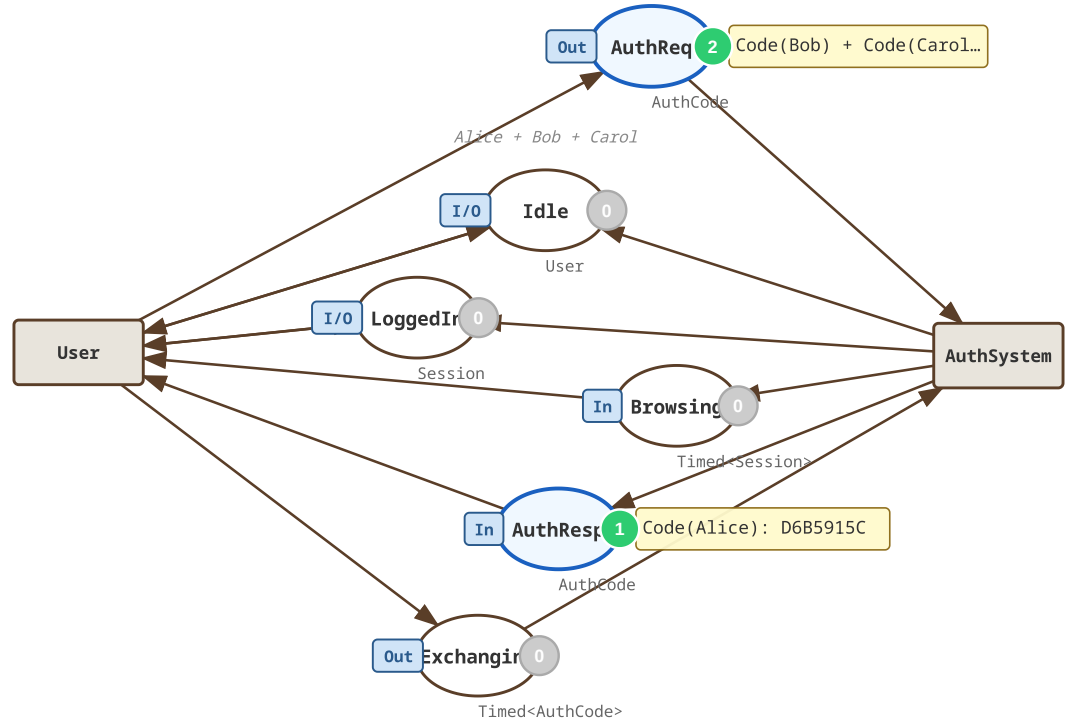

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- Two sub-pages: User & AuthSystem

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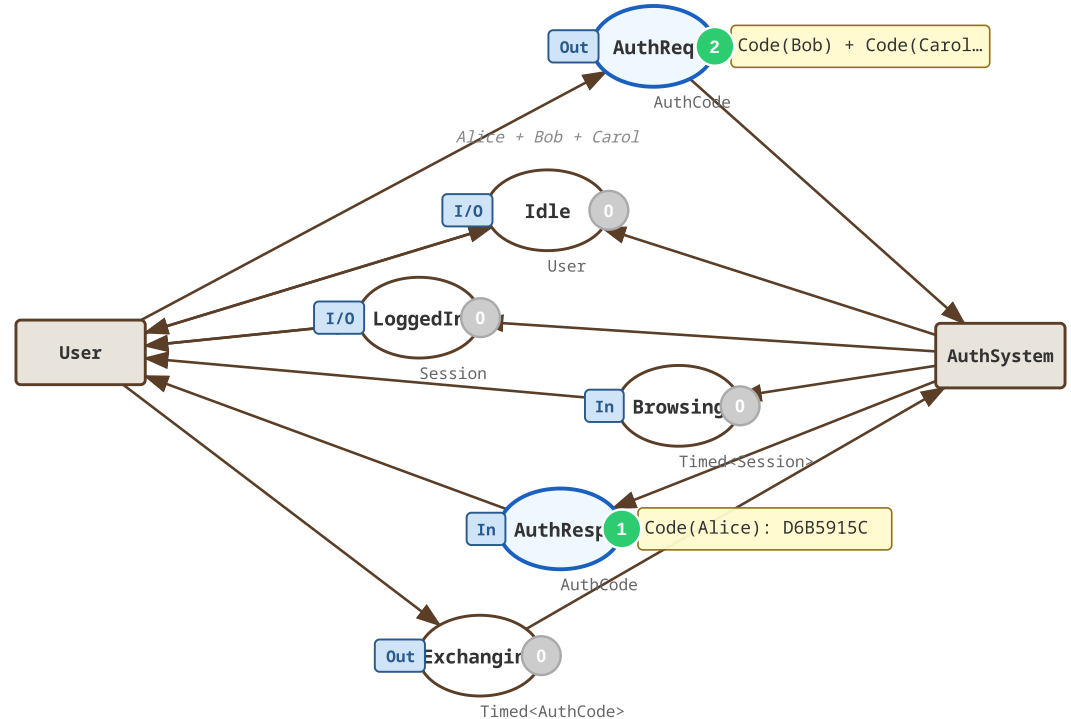

Colour sets are just C# types

Any `notnull, IEquatable<T>` type works — primitives, enums, records, classes.

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public record User(int Id, string Name) {
  public string CsrfState  => $"s{Id}";
  public int    BrowseTime => Id * 5;
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public record AuthCode(User User, string Code, string State) {
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public record Session(User User, int TokenId);
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Places & multisets

```
// inside the model constructor
var authReq = AddPlace<AuthCode>("AuthReq");            // empty initial marking

var idle    = AddPlace("Idle", Multiset.Of(
                new User(1,"Alice"), new User(2,"Bob"), new User(3,"Carol")));

var exchg   = AddTimedPlace<AuthCode>("Exchanging");    // timed place
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var u   = new Var<User>("u");

AddTransition("Login")
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AddTransition("GrantToken")
  .TimedInput(exchg, c).Input(nextTokId, tid)
  .Output(nextTokId, () ⇒ tid.Val + 1)
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  .TimedOutput(expiry,   () ⇒ new Session(c.Val.User, tid.Val), Config.SessionTTL)
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AddTransition("SessionExpire")
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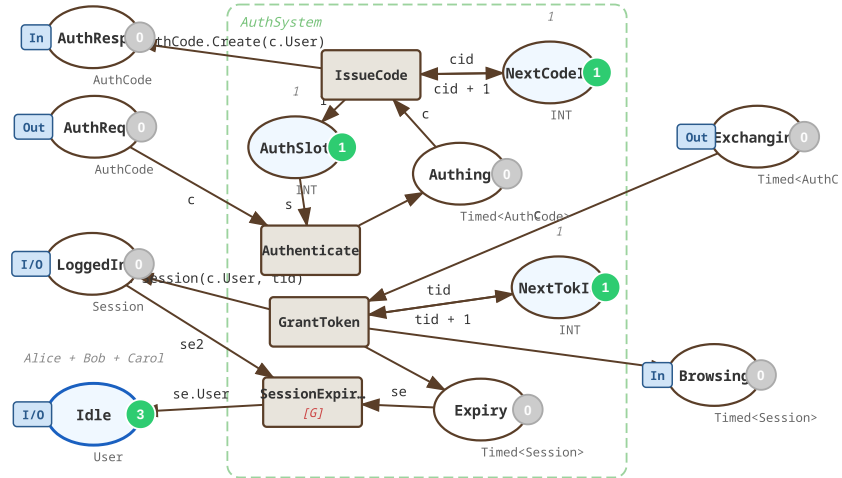
Hierarchical CPNs

```
class UserPage : CpnPage {
  public UserPage(Place<User> idle,
       Place<AuthCode> authReq, ...)
    : base("User") {
    InOut(idle);
    Out(authReq); In(authResp);
    Out(exchg);
    InOut(loggedIn); In(browsing);
    var waiting = AddPlace<User>("Waiting");
    // Login, ReceiveCode, Logout
  }
}
// assembly
AddSubPage(new UserPage(...),       "User");
AddSubPage(new AuthSystemPage(...), "AuthSystem");
```



AuthSystem sub-page: `AuthSlot` limits concurrency, `Expiry` delays timeout.

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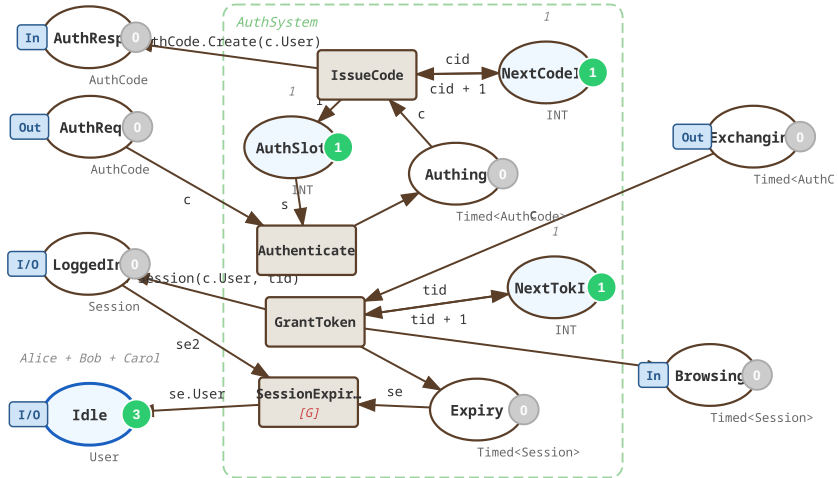

AuthSystem sub-page: `AuthSlot` limits concurrency, `Expiry` delays timeout.

- A sub-page is a `CpnPage` **subclass**

Hierarchical CPNs

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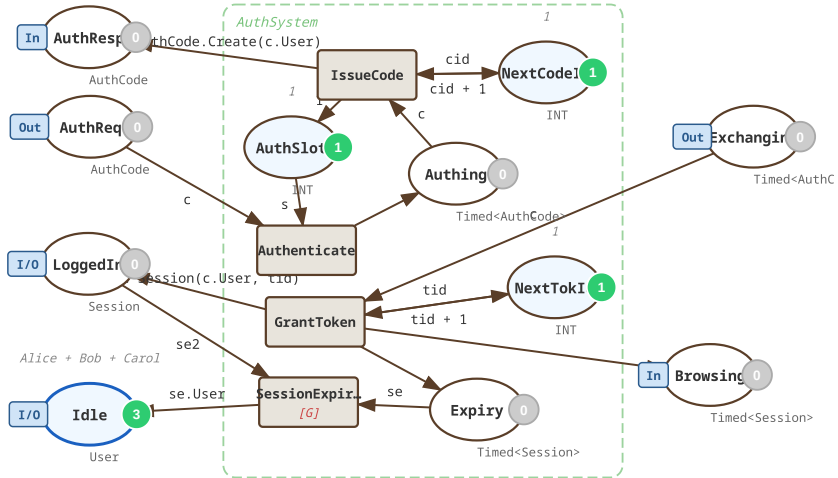

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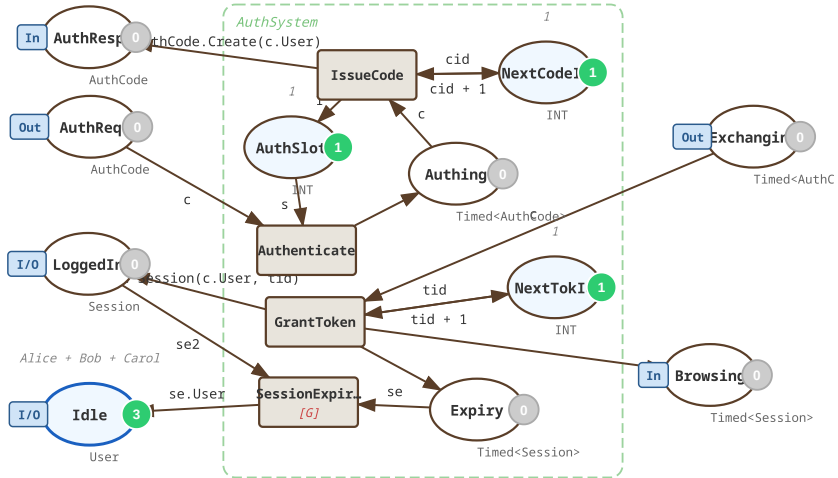

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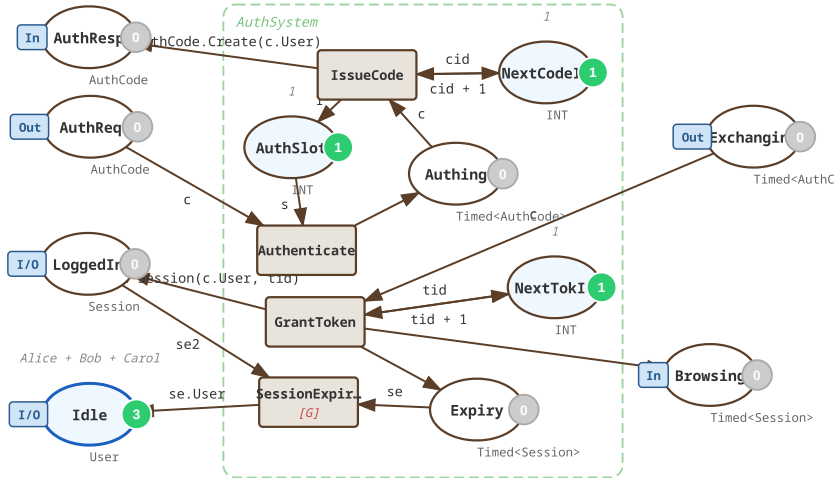

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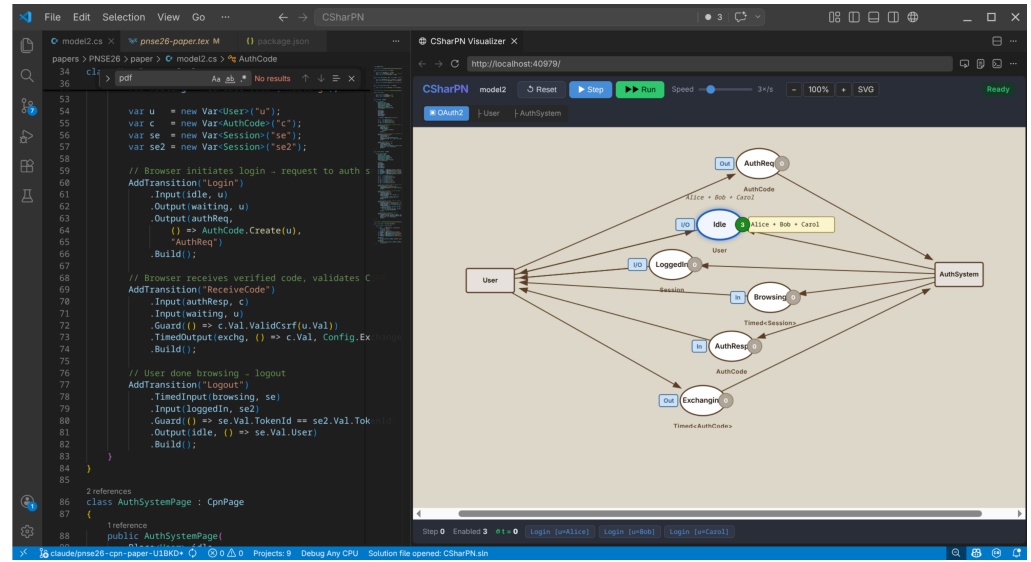

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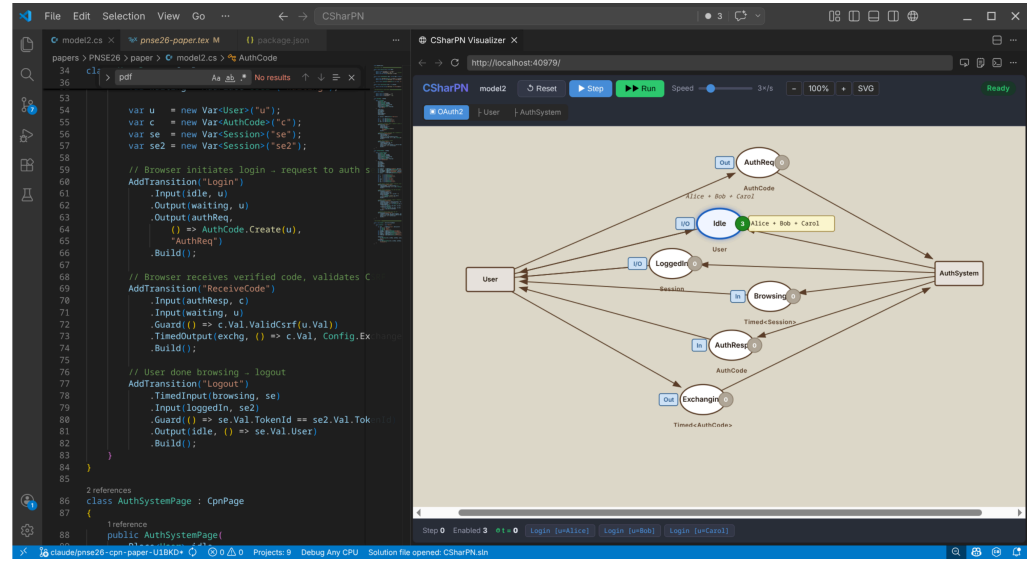

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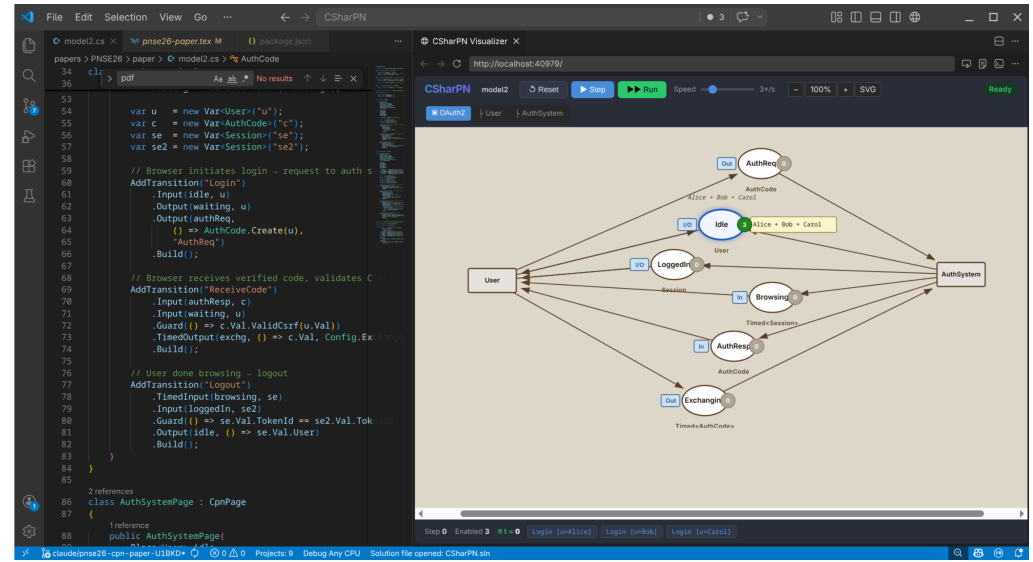
- Page tree — drill into sub-pages



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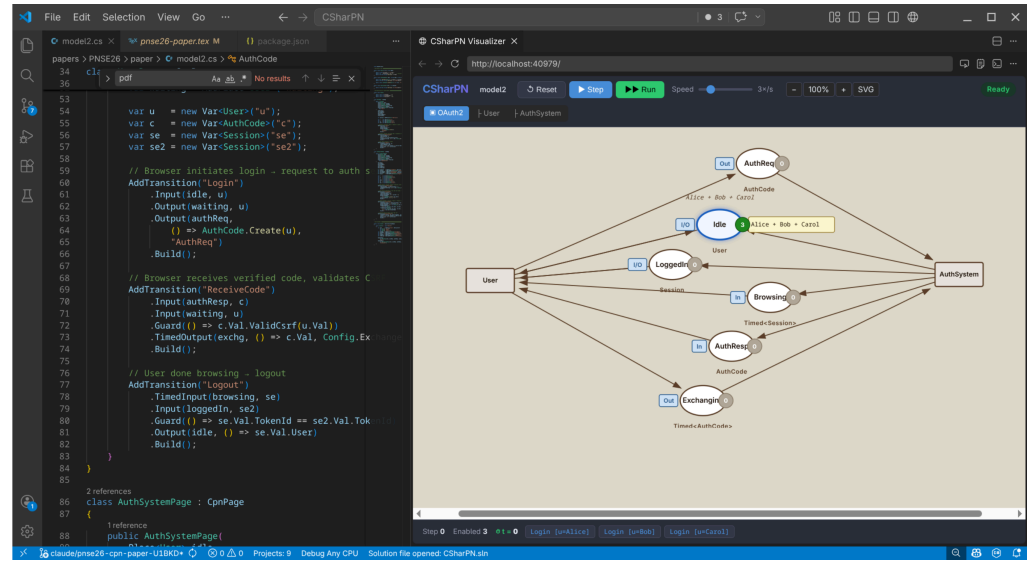

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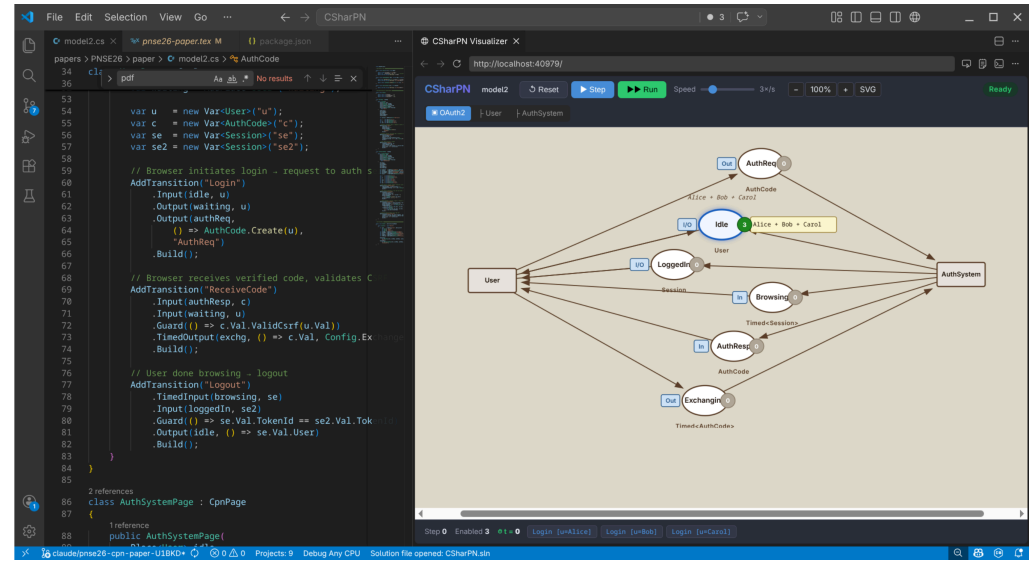

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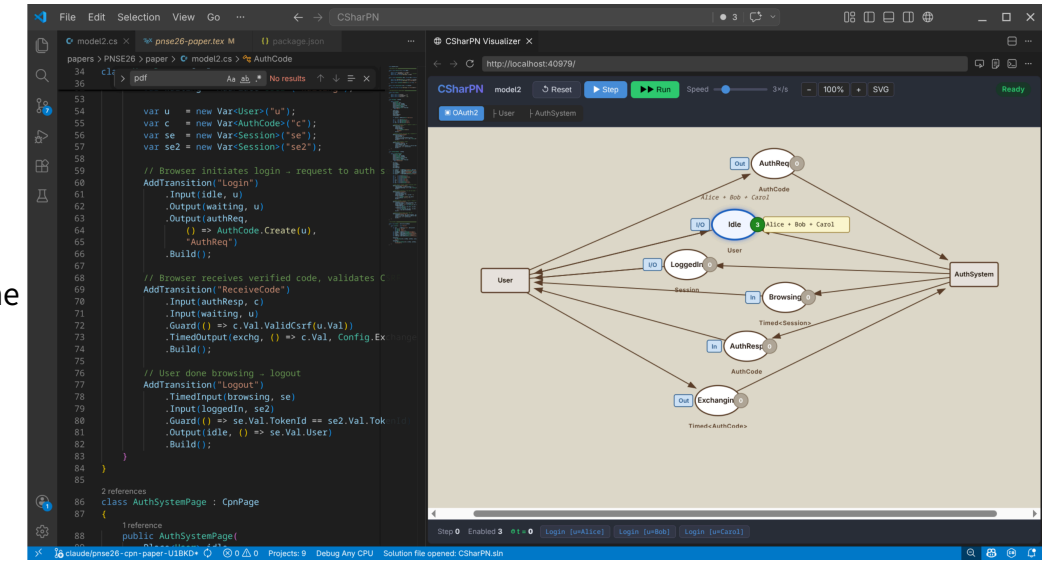

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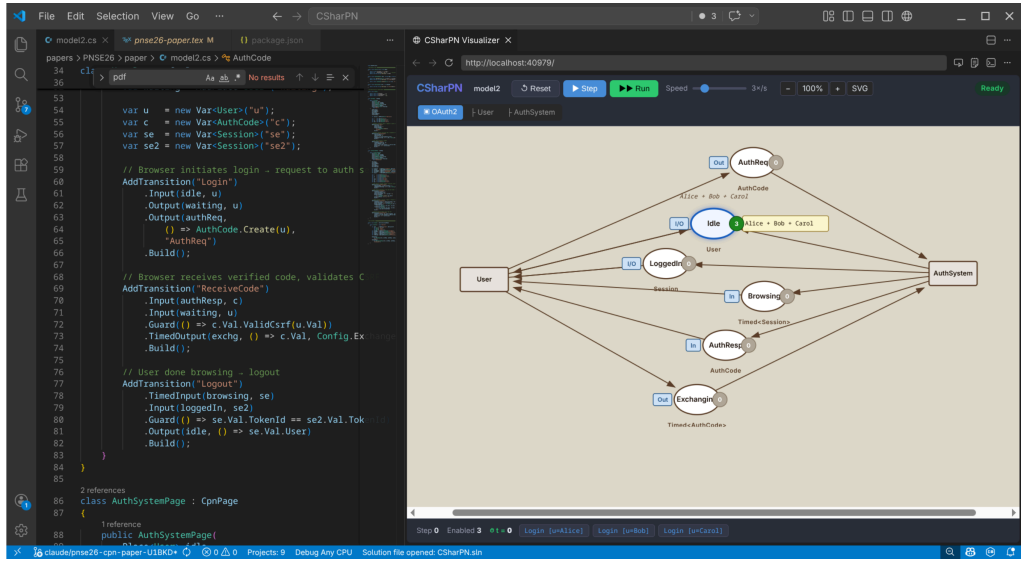

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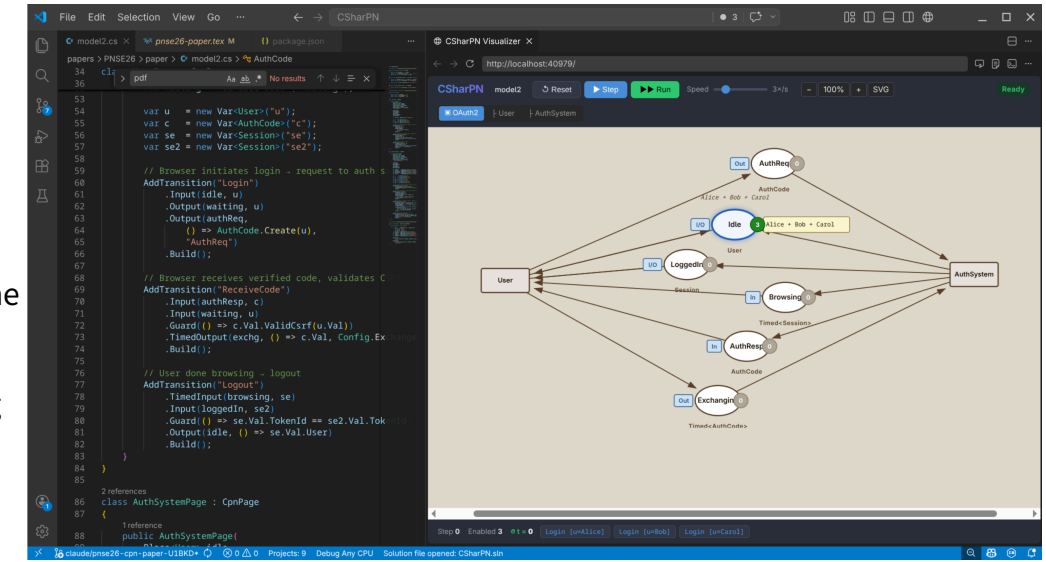

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key insight

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> **No code-generation step.** No generated artefact, no model/implementation sync problem — the gap is zero by construction. (Caveat: only when the deployment target is .NET.)

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> **Takeaway:** when the inscription language is C#, the full power of a mature programming language — type system, libraries, tooling — is available to the CPN modeller at no extra cost.

Thank you

Questions?

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`github.com/simontjell/CSharPN.public`

Simon Tjell · simon@simplesystemer.dk